1.2 MODELS PROPOSED FOR MEMBRANE STRUCTURE

1.2.1. Monolayer Model (1917)

If a lipid molecule containing hydrophilic group is dissolved in a highly volatile solvent and several drops of it are carefully applied to the surface of the water, the lipid spreads out to form a thin mono molecular film. Based on this, Langmuir in 1917 proposed the monolayer model of membranes. In the mono molecular structure, it is found that the hydrophilic portion of each molecule projects in to the water surface and the hydrophobic parts are directed away from the water.

1.2.2. Lipid Bilayer Model or Bimolecular lipid layer:

The model was proposed by Gorter and Grendal in 1925. Lipids from erythrocytes were extracted from erythrocytes and the amount of surface area that lipid would cover when spread over the surface of water was assessed. It was identified that the membrane contained a bimolecular layer of lipids .They found that the polar groups of each molecular layer were directed towards the outside of the bilayer.

1.2.3. Sandwich Model:

Danielli and Davidson proposed the sandwich model in 1935 which was based on the lipid bilayer model. In this model, the plasma membrane is made up of lipid bilayer, sandwiched between the two continuous layers of proteins. The lipid molecules are arranged in an ordered manner and are set at right angles to the surface. The membrane was found to be 75-100 Å thick. In this, the thickness of the protein layer was found to be 20 Å and that of the lipid layer was 35Å .

Drawbacks

This model cannot be applied to all membranes because it reveals definite proportions of lipids and proteins, which has not been true.

1.2.4.Unit membrane Model:

Unit membrane hypothesis of the structure of membrane was proposed in the year 1953 by Robertson. In unit membrane model the protein layers are said to be asymmetrical. The scientist had investigated the structure of a RBC membrane using electron microscope and found that all cellular membranes contains three layers *viz.* two outer layers of the proteins separated by a lighter middle layer of phospholipids. Thus, unit membrane model visualises cell membrane as a trilaminar structure consisting of two dark osmiophilic layers separated by a light osmiophilic layer. The lipid layer was found to be of 35 Å thickness with a dense band of proteins of 20 Å thickness existing on either side of the membrane.Mucoprotein is present on the outer surface and non mucoid protein is present on the inner surface of the RBC membrane.

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